



1. Resuelva las siguientes ecuaciones diferenciales para la función $y(x)$:

a) $y'' - y = 0$

b) $y'' + \omega^2 y = 0$

c) $y^{iv} - y = 0$

d) $y'' + y' + y = 0$

e) $y'' + 2y' + y = 0$

f) $y' = r y(1 - y/K)$

g) $x^2 y'' + 2x y' - 4y = 0$ (puede utilizar el cambio $x = e^u$)

h) $x^2 y'' + 2x y' + 4y = 0$

i) $x^2 y'' + 2x y' + y/4 = 0$

2. Resuelva las siguientes ecuaciones diferenciales para la función $y(x)$:

a) $y'' - y = x$

b) $y'' + \alpha y = \alpha \sin(8x)$

c) $16e^x - 8y - 2y'' = 0$

d) $x^2 y'' + 2x y' - 4y + 4 = 0$

1. Resuelva las siguientes ecuaciones diferenciales para la función $y(x)$:

a) $y'' - y = 0$

La ecuación tiene pol. característico:

$$\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \Rightarrow y(x) = c_1 e^x + c_2 e^{-x} \quad c_1, c_2 \in \mathbb{R}$$

b) $y'' + \omega^2 y = 0$

$$\lambda^2 + \omega^2 = 0 \Rightarrow \lambda = \pm \sqrt{-\omega^2} = \pm i|\omega| \Rightarrow y(x) = c_1 \cos(|\omega|x) + c_2 \sin(|\omega|x) \quad c_1, c_2 \in \mathbb{R}$$

c) $y^{(iv)} - y = 0$

$$\lambda^4 - 1 = 0 \Rightarrow \lambda = \pm 1, \pm i \Rightarrow y(x) = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$$

$c_1, \dots, c_4 \in \mathbb{R}$

d) $y'' + y' + y = 0$

$$\lambda^2 + \lambda + 1 = 0 \Rightarrow \lambda = -\frac{1 \pm \sqrt{3}i}{2} \Rightarrow$$

$$y(x) = e^{-\frac{1}{2}x} \left(c_1 \cos\left(\frac{\sqrt{3}}{2}x\right) + c_2 \sin\left(\frac{\sqrt{3}}{2}x\right) \right) \quad c_1, c_2 \in \mathbb{R}$$

e) $y'' + 2y' + y = 0$

$$\lambda^2 + 2\lambda + 1 = 0 \Rightarrow \lambda = -1 \text{ doble} \Rightarrow y(x) = c_1 e^{-x} + c_2 x e^{-x} = e^{-x}(c_1 + c_2 x)$$

$c_1, c_2 \in \mathbb{R}$

f) $y' = r y(1 - y/K)$

$$\frac{y'}{y(1 - \frac{y}{K})} = r \Rightarrow \int \frac{1}{y(1 - \frac{y}{K})} dy = \int r dx$$

$$\frac{1}{y(1 - \frac{y}{K})} = \frac{a}{y} + \frac{b}{1 - \frac{y}{K}} = \frac{a(1 - \frac{y}{K}) + by}{y(1 - \frac{y}{K})} \Rightarrow 1 = a(1 - \frac{y}{K}) + by$$

$$y = K \Rightarrow 1 = bK \Rightarrow b = 1/K$$

$$y = 0 \Rightarrow 1 = a \Rightarrow a = 1$$

$$\int \frac{1}{y(1 - \frac{y}{K})} dy = \int \frac{1}{y} dy + \frac{1}{K} \int \frac{1}{1 - \frac{y}{K}} dy = \ln|y| - \ln|1 - \frac{y}{K}| =$$

$$\ln\left|\frac{y}{1 - y/K}\right| = rx + C \Rightarrow \frac{y}{1 - \frac{y}{K}} = e^{rx} \Rightarrow y = e^{rx} - y \frac{e^{rx}}{K} \Rightarrow$$

$$y(x) = \frac{e^{rx}}{1 + \frac{e^{rx}}{K}} \quad \text{Sol. des.: } y(1 - \frac{y}{K}) = 0 \Rightarrow y = 0 \text{ o } y = K$$

g) $x^2 y'' + 2x y' - 4y = 0$ (puede utilizar el cambio $x = e^u$)

$$x = e^u \Rightarrow u = \ln x$$

Sea $\tilde{y}(u) = y(x)$

$$y'(x) = \frac{d}{dx} [\tilde{y}(u)] = \frac{d\tilde{y}}{du}(u) \cdot \frac{du}{dx}(x) = \tilde{y}'(u) \cdot \frac{1}{x} = \tilde{y}'(u) e^{-u}$$

$$y''(x) = \frac{d}{dx} \left[\frac{1}{x} \tilde{y}'(u) \right] = -\frac{1}{x^2} \tilde{y}'(u) + \frac{1}{x} \tilde{y}''(u) \cdot \frac{du}{dx} = e^{-2u} (\tilde{y}''(u) - \tilde{y}'(u))$$

$$x^2 e^{-2u} (\tilde{y}''(u) - \tilde{y}'(u)) + 2x (e^{-u} \tilde{y}'(u)) - 4\tilde{y}(u) = 0 \Rightarrow$$

$$e^{2u} e^{-2u} (\tilde{y}''(u) - \tilde{y}'(u)) + 2e^u e^{-u} \tilde{y}'(u) - 4\tilde{y}(u) = 0 \Rightarrow$$

$$\tilde{y}''(u) - \tilde{y}'(u) + 2\tilde{y}'(u) - 4\tilde{y}(u) = 0 \Rightarrow \tilde{y}''(u) + \tilde{y}'(u) - 4\tilde{y}(u) = 0$$

Tronca pol. característico $\lambda^2 + \lambda - 4 = 0 \Rightarrow \lambda = \frac{-1 \pm \sqrt{17}}{2}$

$$\tilde{y}(u) = c_1 e^{\frac{(-1+\sqrt{17})}{2}u} + c_2 e^{\frac{(-1-\sqrt{17})}{2}u}, \quad c_1, c_2 \in \mathbb{R} \Rightarrow$$

$$y(x) = c_1 x^{\frac{-1+\sqrt{17}}{2}} + c_2 x^{\frac{-1-\sqrt{17}}{2}}, \quad c_1, c_2 \in \mathbb{R}$$

h) $x^2 y'' + 2x y' + 4y = 0$

Aplicamos el cambio $x = e^u \Rightarrow u = \ln x$

$\tilde{y}(u) = y(x) \Rightarrow$

$$y'(x) = \frac{d}{dx} (\tilde{y}(u)) = \frac{d\tilde{y}}{du}(u) \cdot \frac{du}{dx}(x) = \tilde{y}'(u) \cdot \frac{1}{x} = \tilde{y}'(u) \cdot e^{-u}$$

$$y''(x) = \frac{d\tilde{y}'(u)}{du} \cdot \frac{du}{dx}(x) e^{-u} + \tilde{y}'(u) \frac{d(e^{-u})}{du}(u) \cdot \frac{du}{dx}(x) =$$

$$\tilde{y}''(u) \cdot \frac{1}{x} e^{-u} - \tilde{y}'(u) e^{-u} \cdot \frac{1}{x} = \tilde{y}''(u) e^{-2u} - \tilde{y}'(u) e^{-2u} = e^{-2u} (\tilde{y}''(u) - \tilde{y}'(u))$$

$$e^{2u} e^{-2u} (\tilde{y}'' - \tilde{y}') + 2e^u e^{-u} \tilde{y}' + 4\tilde{y} = \tilde{y}'' + \tilde{y}' + 4\tilde{y} = 0$$

$$\lambda^2 + \lambda + 4 = 0 \Rightarrow \lambda = \frac{-1 \pm \sqrt{15}i}{2} \Rightarrow$$

$$\tilde{y}(u) = e^{\frac{-1 \pm \sqrt{15}i}{2}u} \left(c_1 \cos\left(\frac{\sqrt{15}}{2}u\right) + c_2 \sin\left(\frac{\sqrt{15}}{2}u\right) \right) \Rightarrow$$

$$y(x) = \frac{1}{x} \left(c_1 \cos\left(\frac{\sqrt{15}}{2} \ln x\right) + c_2 \sin\left(\frac{\sqrt{15}}{2} \ln x\right) \right) \quad c_1, c_2 \in \mathbb{R}$$

$$i) x^2 y'' + 2x y' + y/4 = 0$$

$$\text{sea } x = e^u \Rightarrow u = \ln x$$

$$\tilde{y}(u) = y(x)$$

$$y'(x) = \frac{d\tilde{y}}{du}(u) \cdot \frac{du}{dx}(x) = \tilde{y}'(u) \cdot \frac{1}{x} = \tilde{y}'(u) \cdot e^{-u}$$

$$y''(x) = \frac{d^2\tilde{y}}{du^2}(u) \cdot \frac{du}{dx}(x) \cdot \frac{du}{dx} + \frac{d\tilde{y}}{du}(u) \cdot \frac{d^2}{dx^2}(x) = \tilde{y}''(u) \cdot \frac{1}{x^2} - \tilde{y}'(u) \frac{1}{x^2} = e^{-2u} (\tilde{y}''(u) - \tilde{y}'(u))$$

$$e^{2u} e^{-2u} (\tilde{y}'' - \tilde{y}') + 2 e^u e^{-u} \tilde{y}' + \frac{\tilde{y}}{4} = \tilde{y}'' + \tilde{y}' + \frac{\tilde{y}}{4} = 0$$

$$\lambda^2 + \lambda + \frac{1}{4} = 0 \Rightarrow \lambda = -\frac{1}{2} \text{ doble} \Rightarrow$$

$$\tilde{y}(u) = c_1 e^{-\frac{1}{2}u} + c_2 u e^{-\frac{1}{2}u} = e^{-\frac{1}{2}u} (c_1 + c_2 u) \Rightarrow$$

$$y(x) = \frac{1}{\sqrt{x}} (c_1 + c_2 \ln x) \quad c_1, c_2 \in \mathbb{R}$$

2. Resuelva las siguientes ecuaciones diferenciales para la función $y(x)$:

a) $y'' - y = x$

Probamos con $y(x) = ax$

$$-ax = x \Rightarrow x(a+1) = 0 \Rightarrow a = -1 \Rightarrow y(x) = -x \text{ es sol. particular.}$$

$$\text{Resolvemos } \frac{dy}{dx} = 0 \Rightarrow x^2 - 1 = 0 \Rightarrow x = \pm 1 \Rightarrow$$

$$y(x) = -x + c_1 e^x + c_2 e^{-x} \quad c_1, c_2 \in \mathbb{R} \text{ es la sol. general.}$$

b) $y'' + \alpha y = \alpha \sin(8x)$

Sea $y(x) = a \sin 8x + b \cos 8x$. Como solo tenemos derivadas de orden 0 y 2, basta $y(x) = a \sin 8x$

$$-64a \sin 8x + 64a \sin 8x = \alpha \sin 8x \Rightarrow \sin 8x (-64a + \alpha a) = 0 \Rightarrow a = \frac{\alpha}{\alpha - 64}$$

$$y(x) = \frac{\alpha}{\alpha - 64} \sin 8x \text{ es sol. particular } \forall \alpha \in \mathbb{R} \setminus \{64\}$$

Si $\alpha = 64$, probamos con $x \sin 8x$, $x \cos 8x$ o $x^2 \sin 8x$, $x^2 \cos 8x$...

En particular, $y(x) = -4x \cos x$ es solución particular.

$$x^2 + \alpha = 0 \Rightarrow x = \pm i\sqrt{\alpha} \Rightarrow \text{la sol. general es:}$$

$$y(x) = \begin{cases} \frac{\alpha}{\alpha - 64} \sin 8x + c_1 \cos(x\sqrt{\alpha}) + c_2 \sin(x\sqrt{\alpha}) & \alpha \neq 64 \\ -4x \cos x + c_1 \cos(x\sqrt{\alpha}) + c_2 \sin(x\sqrt{\alpha}) & \alpha = 64 \end{cases} \quad c_1, c_2 \in \mathbb{R}$$

c) $16e^x - 8y - 2y'' = 0$

Probamos con $y(x) = ce^x$, $c \in \mathbb{R}$

$$16e^x - 8ce^x - 2ce^x = 0 \Rightarrow e^x(16 - 10c) = 0 \Rightarrow c = \frac{8}{5} \Rightarrow y(x) = \frac{8}{5}e^x \text{ es sol. particular}$$

$$2x^2 + 8 = 0 \Rightarrow x = \pm 2i \Rightarrow \text{La sol. general es:}$$

$$y(x) = \frac{8}{5}e^x + c_1 \cos(2x) + c_2 \sin(2x) \quad c_1, c_2 \in \mathbb{R}$$

d) $x^2 y'' + 2x y' - 4y + 4 = 0$

Probamos con $y(x) = a \Rightarrow -4a + 4 = 0 \Rightarrow a = 1 \Rightarrow y(x) = 1$ es sol. particular.

Resolviendo la ecuación homogénea, obtenemos la sol. general:

$$y(x) = 1 + c_1 x^{\frac{-1+\sqrt{17}}{2}} + c_2 x^{\frac{-1-\sqrt{17}}{2}}, \quad c_1, c_2 \in \mathbb{R}$$